

5

TOTAL PROJECTS

4

DOMAINS COVERED

10+

TOOLS USED

4 Yrs

TIMELINE SPAN

CHRONOLOGICAL PROJECT TIMELINE

Reverse-chronological order — most recent first

2025

01

PROJECT

Generative AI — NLP Chatbot Development

Jul – Aug 2025

FincruX Technologies LLP · Artificial Intelligence & NLP

TYPE	DURATION	SECTOR	STATUS
Professional	2 Months	AI / FinTech	Completed

Tools & Technologies: Python · Hugging Face Transformers · Gemini API · Prompt Engineering · Custom Typing-Animation UI

Key Contributions & Findings

- Built 3–5 NLP chatbots including an English–French translator via Hugging Face pre-trained models and a poetic generation bot powered by the Gemini API.
- Achieved 40–60% higher user engagement by designing a custom typing-animation UI that simulated human-like, real-time conversational responses.
- Engineered 2+ persona-driven lyrical chatbots with advanced prompt engineering using 15+ contextual constraints to maximise response accuracy and tonal consistency.
- Optimised multi-turn context retention through structured prompt memory management, preventing context drift across extended conversations.
- Demonstrated cross-framework proficiency by concurrently leveraging Hugging Face open-source models alongside Google Gemini's generative API capabilities.

OUTCOME 40–60% engagement lift · 5 chatbots deployed · 15+ prompt constraints engineered

02

PROJECT

Business Analysis — Power BI Dashboards & Strategic Diagnostics

Feb – Apr 2025

FincruX Technologies LLP · Business Analytics & Strategy

TYPE	DURATION	SECTOR	STATUS
Professional	3 Months	Consulting	Completed

Tools & Technologies: Power BI · DAX · SQL · Excel · MECE Framework

Key Contributions & Findings

- Built 5 Power BI dashboards (TreeMap, Funnel, Gauge, Line) using DAX measures and SQL-sourced data, enabling real-time KPI tracking across cross-functional teams.
- Improved leadership decision-making speed by consolidating fragmented operational data into drill-down dashboards with auto-refreshing metrics.
- Diagnosed a critical 62% employee-cost vs. under 21% customer-base structural imbalance via MECE-based analysis of the company's cost-revenue architecture.
- Formulated targeted acquisition recommendations and operational efficiency strategies projected to drive 15%+ improvement in net profit margins.
- Collaborated with cross-functional stakeholders to validate dashboard logic and ensure full business alignment before final deployment.

OUTCOME 5 dashboards built · 62% cost anomaly diagnosed · 15%+ margin improvement recommended

2022

TYPE	DURATION	SECTOR	STATUS
Academic	1 Month	Capital Markets	Completed
Tools & Technologies: RStudio · Excel · CAPM Model · Beta Regression · NSE Historical Data			
Key Contributions & Findings			
- Applied CAPM to evaluate expected vs. actual returns for all constituent stocks of the Indian IT Nifty Index using NSE historical price and return data. - Computed beta, expected return, and alpha for each stock to quantify systematic risk and identify pricing anomalies across the index. - Found 90% of IT sector stocks consistently exceeded CAPM-predicted returns, revealing persistent model underestimation in Indian technology equities. - Identified Mphasis and Mindtree as significantly undervalued — actual returns were 701% and 268% above CAPM benchmarks respectively, representing strong alpha signals. - Contributed empirical evidence for anomaly-driven return deviations in emerging market equities, challenging classical CAPM assumptions in the Indian context.			
OUTCOME	90% of stocks beat CAPM · Mphasis +701% vs benchmark · Mindtree +268% vs benchmark		

TYPE	DURATION	SECTOR	STATUS
Academic	3 Months	Env. Economics	Completed
Tools & Technologies: RStudio · Excel · OLS Regression · GDP Modelling · EV & Alt-Fuel Sector Data			
Key Contributions & Findings			
- Analysed 16 years of U.S. macroeconomic and EV industry data to quantify the economic impact of decarbonisation on GDP growth, employment, and energy transition. - Produced a deep-dive analysis of the \$28.24 Bn U.S. EV industry covering production trends, policy incentives, and market growth trajectories across the study period. - Visualised EV adoption curves and alternative fuel usage in aviation using RStudio, providing a multi-sector view of the energy transition's economic footprint. - Built an OLS regression model linking decarbonisation variables to U.S. GDP per capita — near-perfect model fit: $R^2 = 0.99$, F-statistic = 385.43. - Extended research scope to include the aviation sector's alternative fuel adoption, broadening the study's macroeconomic relevance and policy applicability.			
OUTCOME	$R^2 = 0.99$ · F-stat = 385.43 · \$28.24 Bn industry analysed · 16 years of data		

05 PROJECT	Agricultural Production — Foundation of the Indian Economy			Nov 2021
	Data Analysis · Hansraj College, University of Delhi			
TYPE Academic		DURATION 1 Month	SECTOR Agri. Economics	STATUS Completed
Tools & Technologies: Excel · Govt. Survey Data (MOSPI / ICAR) · Statistical Trend Analysis				
Key Contributions & Findings				
▶ Analysed Indian foodgrain production data spanning two decades to map structural trends, seasonal patterns, and policy-driven productivity shifts. ▶ Conducted multi-source data collection via primary surveys, expert interviews, and government databases (MOSPI, ICAR) for comprehensive sectoral coverage. ▶ Quantified the Green Revolution's impact — national agricultural output doubled to 1,160 lakh tonnes — establishing it as the critical driver of India's food security. ▶ Examined linkages between production volumes, rural employment rates, and India's broader GDP contribution across the study period. ▶ Delivered policy insights on irrigation expansion, crop diversification, and technology adoption as levers for sustaining future productivity growth.				
OUTCOME		100% output rise quantified · 1,160 lakh tonnes documented · 20+ years of data analysed		

TECHNICAL TOOLKIT & CERTIFICATIONS

Financial Modelling & DCF	Power BI & DAX	Python (NLP / Data Analysis)	SQL & Databases
RStudio & Econometrics	Hugging Face / Gemini API	Excel (Advanced)	Monte Carlo Simulation
CAPM & Equity Research	Prompt Engineering	Variance Analysis	OLS Regression
NISM Research Analyst 91.5% · 2025	SQL for Data Science UC Davis / Coursera · 98.26%	Python for Data Analysis IBM · 2023	Intro to Corporate Finance UPenn · 2023
Investment Banking Course 365 Careers · 2024			